

Beating Round 2

In class, someone asked about how beating works when there are multiple signals as opposed to just two. I said that in general, there isn't any special pattern, and you will just get kind of random loudening/softening. That's indeed generally true, but if you pick your beating signals very, very carefully, then you can get dramatically different behavior. We'll look a little bit at this.

1. Two-signal Beating

If we have two closely-spaced in frequency sine/cosine waves, we saw that we get *beating*. Instead of hearing two slightly separate notes, we hear one note that gets louder and softer. Mathmatically, that's because

$$\cos(2\pi(\nu_0 + d\nu)t) + \cos(2\pi(\nu_0 - d\nu)t) = 2 \cos(2\pi\nu_0 t) \cos(2\pi d\nu t) \quad (1)$$

We have a fast sine wave $2 \cos(2\pi\nu_0 t)$ that our ear hears as a pitch that gets modulated by $\cos(2\pi d\nu t)$. If $d\nu$ is small enough (less than about 20 Hz) then our ears hear this as the main pitch at ν_0 getting louder and softer as the $d\nu$ cosine goes between 1 and 0 (and -1 and back to 0).

What we have written down assumes that our two notes start at exactly the same time. Generally this won't be true, and one note will have started slightly later than the other, so what we *really* hear is

$$\cos(2\pi(\nu_0 + d\nu)(t - dt)) + \cos(2\pi(\nu_0 - d\nu)t) \quad (2)$$

Equation 2 looks a lot like Equation 1, except for the dt . That dt says that the first note started just a little bit before the second note. If you go through the (now much more painful!) math, you'll discover that nothing really changes. *When* the beats happen will change, but the beating pattern itself remains the same.

2. Four-note Beating

Now let's look at what happens when we have four notes going at the same time. We'll start with the notes starting together ("in phase" is the usual technical term), with frequencies $\nu - 2d\nu, \nu - d\nu, \nu + d\nu, \nu + 2d\nu$. We can sum the $\pm d\nu$ pair and the $\pm 2d\nu$ pair to get:

$$2 \cos(2\pi\nu t) \cos(2\pi d\nu t) + 2 \cos(2\pi\nu t) \cos(2\pi 2d\nu t) = 2 \cos(2\pi\nu t) (\cos(2\pi d\nu t) + \cos(4\pi d\nu t)) \quad (3)$$

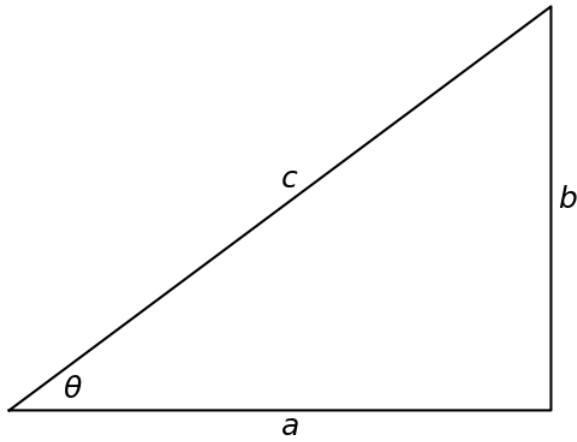


Fig. 1.— Plot of a right triangle. If a, b, c are the length of the corresponding sides in the figure, then we have $\cos(\theta) = a/c$ and $\sin(\theta) = b/c$.